

Aegis-Class Rover

Lunar Surface Mobility Platform

Architecture Reference Document | Rev B — September 2026 | Not for Hardware Commitment

1. Executive Summary

The Aegis-Class Rover is a pressurized, long-duration lunar surface mobility platform designed to support sustained operations at the lunar south pole and other high-value exploration sites. It combines extended-duration crew habitation (30–60 days), integrated power and thermal architecture optimized for low solar elevation, suitport-based dust boundary control, radiation storm shelter capability, and a NASA cFS-based flight software and crew operations console.

ASI develops the vehicle-defining systems: the pressurized crew module, the integrated power and thermal architecture (roof radiator and the ASI node power/thermal interface), and the flight software and operations console. The mobility chassis is sourced rather than rebuilt — drawn from the maturing open-architecture lunar rover market — with ASI specifying the interface and integrating the platform.

The vehicle is architected as a surface infrastructure element, not a short-sortie exploration rover. It is intended to operate as part of a distributed logistics network including orbital habitats, surface outposts, LUNET utility nodes, and ISRU systems.

2. Mission Architecture & Operational Concept

Primary Mission Roles

- Polar water prospecting support
- Surface logistics transport
- Mobile command for construction and field operations
- ISRU equipment towing and servicing
- Emergency shelter and crew lifeboat

Operational Regime

Parameter	Value
Nominal crew	2–4
Emergency crew	Up to 6 (short-duration)
Mission duration	30–60 days
Range	~76 km per battery charge on firm ground (46 km soft); 500 km via LUNET recharge (energetic, not geometric); ~27 km/day sustained on solar
Cruise speed	~11 km/h firm, ~6 km/h soft, at 1.5 kW continuous per wheel
Maximum speed	~15 km/h, terrain and hazard-detection limited

3. Design Philosophy — Constraint Hierarchy

The rover geometry is derived from a strict constraint chain, ensuring traceability from crew habitability to vehicle mass and stability:

- 1 Interior standing height (2.1 m clear)
 - 2 Exterior hull dimensions
 - 3 Center of gravity (CG) height
 - 4 Track width and static stability requirement
 - 5 Roof footprint for power/thermal systems
 - 6 Radiator and solar array sizing
 - 7 Electrical and thermal system capacity
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4. Dimensional & Stability Parameters

Exterior Geometry

Parameter	Value
Overall length	~10.0 m
Overall width	4.7 m (4.1 m hull over a 4.0 m track, 0.7 m wheels)
Track width (center-to-center)	4.0 m
Wheelbase	6.5 m
Hull section	Flattened cylinder, 4.1 × 2.7 m exterior (3.9 × 2.5 m interior), 1.4 m flats, 1.35 m side-wall radius
Height to hull roof	3.65 m
Height to roof radiator top	3.80 m
Ground clearance	~0.50 m

Stability Metrics

Parameter	Value
CG height	~1.64 m above ground (group-height estimate, TBR)
Static Stability Factor (SSF)	~1.22 — just above the 1.2 floor; track or battery-bay height is the lever
NASA target (crewed)	≥ 1.2

The 4.0 m track width provides margin for dynamic loads, slope traversal, and cargo asymmetry.

5. Mass Budget (Rev B, bottom-up)

Group	Mass (kg)
Pressure hull (flattened cylinder), deck, racks, outer layer, mounts	2,231
Mobility platform — partner allocation (frame, 6 stations, steering, brakes)	1,226
Battery pack 100 kWh LFP, PMAD, harness	1,196
Fuel cell, reactants for 5 shadow days, COPV tanks	428
Roof radiator, node interface plate, wings, fluid loop	280
ECLSS hardware, O ₂ and N ₂ with tanks	747
Water 700 kg (200 consumable + 500 shelter) and bladders	840
Interior outfit, avionics, sensors, comms	700
Crew (3 × 80 kg), 2 EVA suits, tools	590
Food, clothing, spares, medical, science allocation	773
Growth margin, 20 % on dry hardware	1,504
Gross vehicle mass	~10,489

Chassis, suspension, and drivetrain mass is the platform partner's allocation against the integrated-vehicle budget. The Rev A figure of ~5,400 kg carried no food, water, oxygen or reactant contents, no growth margin, and a cylinder's hull mass on a box-shaped cabin; the Rev B budget is built line by line with the bases stated, and the hull comes from a cross-section trade that replaced the 3.6 × 2.3 m box with a 4.1 × 2.7 m flattened cylinder at about half the hull mass for the same width at shoulder height. Line items are held in the Rev B model set.

6. Mobility Platform Interface

The mobility chassis — suspension, drivetrain, wheels, and body structure — is sourced from a third-party platform partner drawn from the maturing open-architecture lunar rover market. ASI specifies the interface and integrates the platform; the reference configuration below states the operational envelope and interface requirements the crew module brings to any candidate platform. Specific chassis selection and final parameter values follow once a platform partnership is established around this envelope.

Reference Structural Frame

- Rigid metal or composite chassis (partner-defined)
- Multi-axle geometry (6-wheel reference configuration)
- Independent terrain-following suspension
- Wide stance for lateral slope stability

Reference Wheel System

- Non-pneumatic metal-mesh flex wheels
- Designed for regolith abrasion and thermal cycling
- Ground-pressure target ≤ 7 kPa for south-pole regolith

These figures represent design-baseline expectations consistent with currently demonstrated lunar mobility platforms and serve as the integration starting point for partner discussions.

7. Flight Software & Crew Operations Console

The rover flight software is built on NASA's Core Flight System (cFS) — the same flight-heritage framework used across NASA missions. Mission-application source is identical across simulation, bench, and flight targets; only the hardware abstraction layer (HAL) changes between environments, so verification performed in simulation carries forward rather than being rebuilt for each phase.

Development status: The flight software runs end-to-end in a desktop simulation — a real cFS workspace, all 16 applications, with autonomous FDIR verified live across the full safing chain. *Bench and on-vehicle targets share the same source through the HAL; no flight hardware is committed.*

Parameter	Value
cFS applications	16 (cFE core + 9 standard cFS + 7 Aegis mission apps)
FDIR safing modes	5 — NOMINAL → THERM_SAFE → ECLSS_SAFE → LOADSHED → SHELTER
Navigation update rate	20 Hz (aegis_nav navigation & odometry)
Telemetry protocol	CCSDS (standards-compliant)
Console link	WebSocket bridge (CCSDS-UDP ↔ browser)
Runtime	cFE — Core Flight Executive, NASA flight heritage

Autonomous Fault Detection, Isolation & Recovery

- Continuous watchpoint monitoring across thermal, life support, electrical, and radiation domains
- Autonomous transition through the safing chain without requiring crew action
- Each mode a documented operational posture (entry conditions, reconfiguration, crew advisories)
- Crew override authority preserved at all levels

Autonomous Waypoint Navigation

- Waypoint catalog and guidance state machine for autonomous traverse
- Reachability gated by live state-of-charge, vehicle load, and multi-hop LUNET reachability — energetic, not geometric range
- Planner accounts for available recharge stops along candidate routes

- LIDAR + visual SLAM terrain classification; manual, semi-autonomous, and fully autonomous modes; return-to-base; real-time hazard detection

Crew Operations Console

- Browser-based real-time interface for situational awareness and command authority
- Pages: summary status, navigation / waypoints, power / thermal, atmosphere / ECLSS, pilot view, contingency recovery
- Identical across simulation, bench, and flight — consistent operator training and procedure validation

The same flight software and console are designed to operate across whichever mobility chassis the crew module is paired with; the integration spine is consistent across platform partnerships.

8. Command Module & Habitability

Interior Dimensions

Parameter	Value
Interior length	8.0 m
Interior width	3.9 m at mid-height; 3.1 m at deck and shoulder height; 2.1 m at full headroom (aisle)
Interior height	2.1 m clear
Pressurised volume	~66 m³ (59 m³ above the deck to 2.1 m, 7 m³ cove under the deck)
Net habitable volume	~30–35 m³ — about 10 m³ per crew for 60 days, below the ~25 m³ long-duration guideline; a known compromise

Functional Zones

- **Zone A:** Forward Command
- **Zone B:** Central Workspace / Galley
- **Zone C:** Crew Quarters / Medical
- **Zone D:** Systems Bay
- **Zone E:** Suitport Vestibule

Cabin Atmosphere

Parameter	Value
Total pressure	56.5 kPa (8.2 psia) — NASA exploration atmosphere
O ₂ partial pressure	19.2 kPa
O ₂ fraction	34% — under the 36% flammability line, so the NASA-STD-6001 materials database applies
Temperature	18–22°C
Relative humidity	40–55%

Reduced pressure lowers structural mass and reduces EVA prebreathe requirements.

9. Life Support & Consumables

ECLSS Components

- Dual-bed CO₂ scrubber
- Trace contaminant control
- Humidity condensation loop
- Water recovery (85% nominal)
- High-pressure O₂/N₂ storage

Water Budget (3 Crew, 60 Days)

Category	Gross Demand (kg)
Drinking	360
Food rehydration	90
Hygiene	90
Medical / contingency	36
Total gross demand	576

With 85% recovery, net makeup requirement ≈ 86 kg. Vehicle carries 700 kg at mission start: 200 kg consumable fill plus 500 kg in the storm-shelter bladders, which doubles as reserve.

10. Suitport & Dust Management

EVA Architecture

- External suitport docking
- No full-cabin depressurization required
- Faster ingress/egress
- Reduced atmosphere loss

Vestibule Systems

- Dedicated HEPA filtration
- Electrostatic dust precipitation
- Sticky mat threshold control
- Cyclonic vacuum
- Mandatory wipe-down protocol

Fallback capability: traditional airlock mode.

11. Radiation Protection

Solar Particle Event Shelter

- Deployable water-wall shielding
- 700 kg of water in deployable bladders around a three-person huddle: ~7 cm on five faces, plus food, tanks and racks; 10 cm would need ~1,044 kg
- Underfloor water tanks provide bottom shielding

Estimated severe SPE dose reduced from lethal exposure range to survivable emergency exposure range.

Galactic Cosmic Radiation

- 0.3–0.6 mSv/day baseline
 - 60-day exposure within occupational limits
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12. Power Architecture

Electrical Loads

Mode	Power
Cruise (11 km/h, 1.5 kW per wheel)	~11,200 W incl. hotel (1.04 kWh/km at 10.5 t)
Station-keeping	~2,170 W
Peak	~8,890 W

Generation Strategy

Source	Capability
Roof array	None in Rev B: a flat 20 m² array made ~0.8 kW at 6° and blocked the radiator; a sun-tracking mast array for driving is TBD
Deployable wing arrays (10 m², sun-tracking when parked)	~3.5 kW combined; the only solar source until the mast array is sized
Fuel cell	3 kW; reactants and tanks for 5 shadow days (~430 kg)
Battery storage	100 kWh, 4 × 25 kWh LFP; one traverse day
Outpost recharge	LUNET node, logistics-based

13. Thermal Architecture — Roof Radiator

Roof Radiator, Unshaded

Parameter	Value
Area	6.0 m², horizontal, white coating, on the hull roof
Operating temperature	310 K
Net rejection, clean (ϵ 0.88, α_s 0.12, 6° sun)	~444 W/m² — ~2,662 W
Net rejection, dusty end of mission (ϵ 0.75, α_s 0.25)	~357 W/m² — ~2,143 W (sizing case)
Solar load on a horizontal surface at 6°	~142 W/m² incident, ~17 W/m² absorbed clean
Roof interface	Node power and thermal connection plate — the ASI standard shared with the hopper

Roof array tier: trade result

Rev A carried a 20 m² solar array 0.4 m above the radiator to shade it. At polar sun angles the shade saves almost nothing, because the sun grazes a horizontal surface, while the array blocks the radiator's view of space. With the documented view factor of 0.30 the shaded panel rejected ~302 W/m² against ~444 W/m² unshaded; the view factor for a full-coverage array at 0.4 m is about 0.73, which takes the shaded figure to ~75 W/m². The shadow itself needs a 3.8 m overhang at 6° to cover the panel, so the low sun passed under the array regardless. The concept is kept for high-sun-angle assets and is described as such on its own page.

Mass & CG Impact

- Radiator panel and fluid loop: ~140 kg; interface plate ~25 kg
 - Roof array tier deleted: -0.30 m total height, CG lowered
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14. Development Phasing

Current Status

- cFS flight software and crew console running end-to-end in full software simulation
- Autonomous FDIR verified live across the safing chain
- Mobility platform partner selection in progress

Phase 1

- Uncrewed demonstrator
- Roof radiator and node-interface validation
- Dust and suitport testing
- Flight software to bench and hardware-in-the-loop targets through the HAL

Phase 2

- Crewed 7-day sortie capability
- Partial closed-loop ECLSS

Phase 3

- Full 30–60 day mission capability
 - ISRU integration and surface logistics chain
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15. Programmatic Position

The Aegis-Class Rover is designed to:

- Bridge short-duration sorties and permanent surface habitats
- Enable distributed surface industry

- Operate as part of a broader cislunar logistics architecture
- Provide safe, stable, long-duration surface mobility without requiring immediate fixed megastructure deployment
- Concentrate ASI development on the crew module, integrated power/thermal, and flight software — the vehicle-defining systems — while sourcing the mobility chassis from the open-architecture lunar rover market